### Updated Project Brief: Introvert (Side-Loaded Mesh)

**Introvert** is a decentralized, privacy-first communication platform that utilizes a **Hybrid Mesh architecture** to overcome the background connectivity limitations of mobile operating systems. By bypassing centralized app stores, the project implements a direct **Solana-based token economy** that incentivizes users to provide infrastructure for the network.

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### Core Project Aim

The mission is to create a secure, serverless messaging and VoIP environment where availability is maintained by a mutualistic relationship between peers. Stable nodes (Anchors) provide zero-knowledge storage for mobile nodes (Ephemeral), and this service is powered by a native **Introvert Token** on the **Solana** blockchain.

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### Technical Execution Strategy

- **Platform & Distribution:**
  - **Side-Loading Strategy:** Bypassing Apple/Google stores via a **Capacitor-based** mobile app and direct Linux/Android distribution.
  - **Native Rust Core:** A high-performance engine linked to the UI via a **C-ABI/FFI bridge**.
- **P2P Networking & Identity:**
  - **Persistent Identity:** Secure **Ed25519** keypairs that function as both a P2P node ID and a **Solana Wallet**.
  - **Advanced Transport:** Utilizing libp2p with **QUIC** for roaming stability and **WebRTC** for NAT hole-punching.
- **Data & Consistency:**
  - **Local-First Storage: SQLCipher** encrypted databases with HKDF-derived keys.
  - **Conflict Resolution: CRDTs** for group management and **Lamport/Vector Clocks** for causal message ordering.
- **Incentive Economy (Phase 6):**
  - **Token Model:** A native SPL Token on Solana used to "meter" network usage and reward infrastructure providers.
  - **Proof of Storage:** Anchor nodes earn tokens by providing "Store and Forward" (mailbox) services for offline mobile peers.

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### Phase-by-Phase Roadmap

| Phase | Milestone | Focus |
|---|---|---|
| 1 | **Core Mesh** | Rust workspace, libp2p swarm, and Ed25519 persistence. |
| 2 | **State Layer** | SQLCipher integration, CRDTs, and causal message ordering. |
| 3 | **Anchor Service** | Uptime heuristics and zero-knowledge "Mailbox" protocols. |
| 4 | **VoIP Bridge** | C-ABI interface and stateful **Opus** audio frame management. |
| 5 | **Capacitor UI** | Side-loaded app with a "White Tech" glassmorphic aesthetic. |
| 6 | **Solana Integration** | SPL Token minting, wallet derivation, and incentive logic. |

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**Planned Technology Stack**

- **Systems:** Rust (Engine), TypeScript (Capacitor/UI).
- **Networking:** libp2p (Kademlia DHT, QUIC, WebRTC).
- **Cryptography:** Noise Protocol, Double Ratchet (Signal), Ed25519.
- **Blockchain:** Solana (SPL Token), solana-sdk.
- **Storage:** SQLite/SQLCipher.
- **Audio:** Opus Codec.